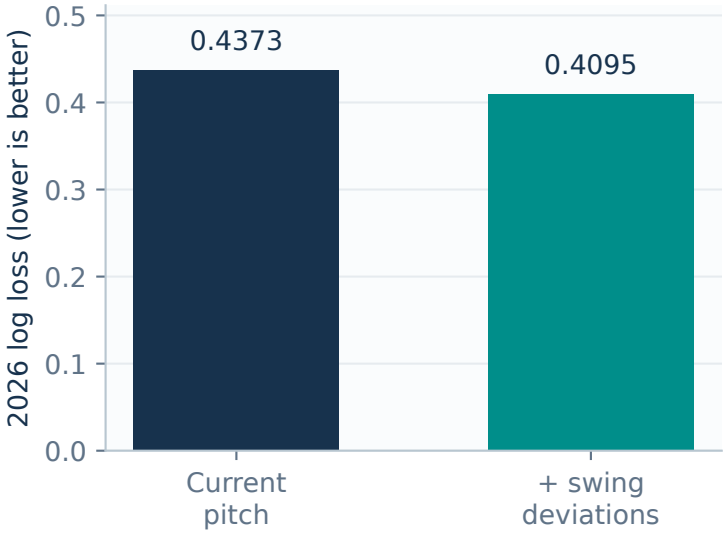
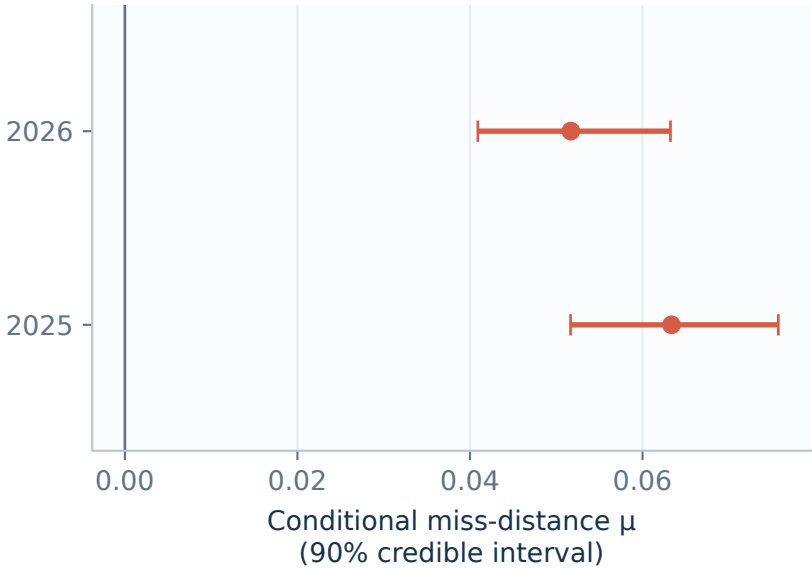


Outcome validity and tracking selection

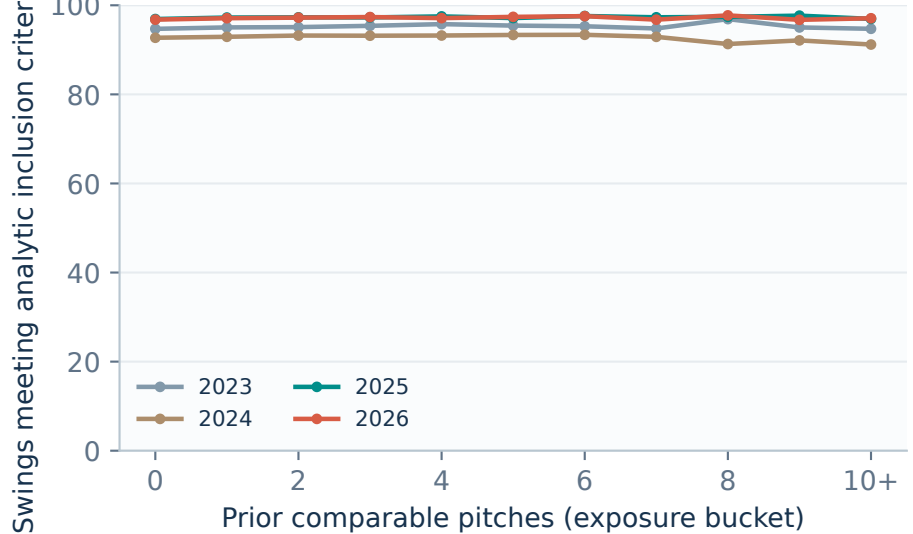
Held-out whiff discrimination



Only tracked whiffs



Tracking inclusion by exposure



Whiff classification uses contemporaneous mechanics, not a pre-pitch forecast. Miss distance is conditional on tracked whiffs; contacts are missing, not zero. Miss-distance units: 2025 SD of $\log(1 + \text{distance})$ per $\log(1 + \text{exposure})$. No unconditional error-learning claim follows from this selected endpoint.